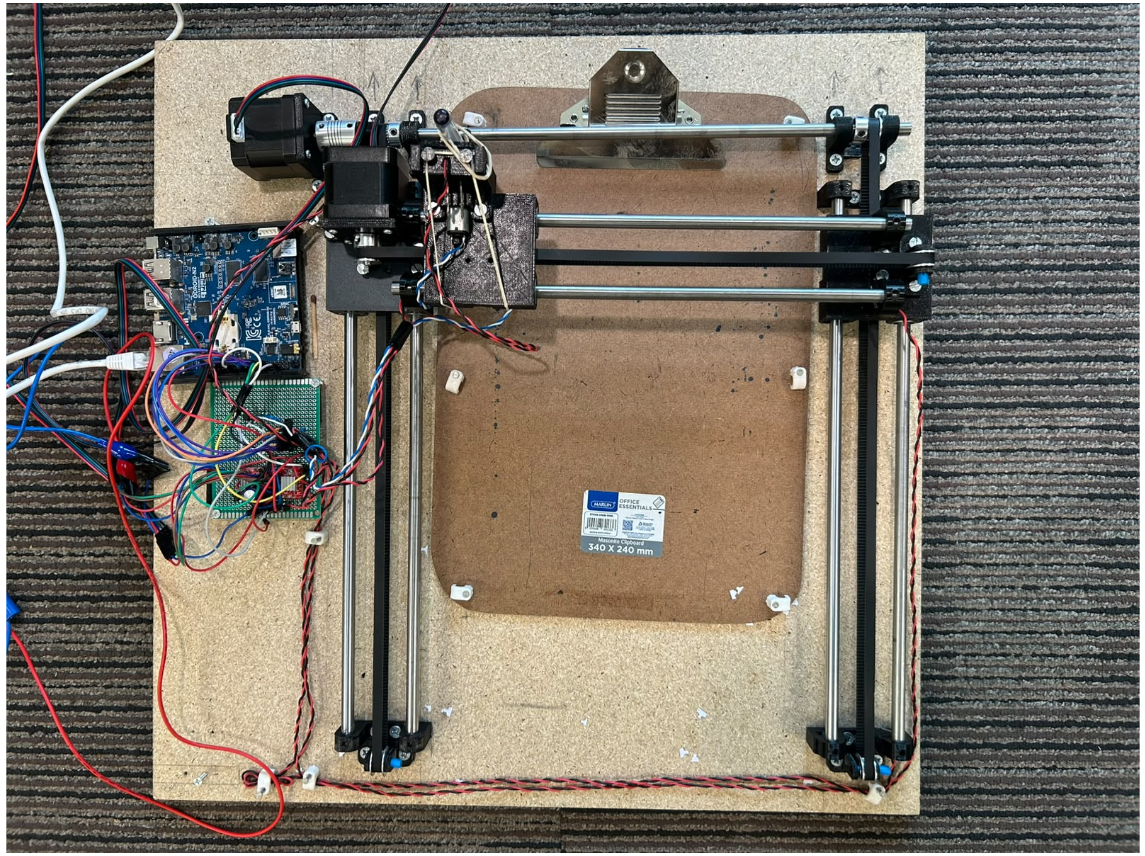
Handwriting robot

Recognising a page and reproducing an individual writing style

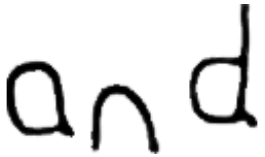


Actual handwriting samples

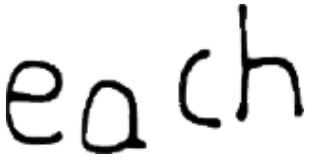
Writer 150



Writer O



Writer Y



11 stored styles. One calibrated pen gantry.

Full system operation

An offline preparation stage supports two operating paths

PREPARE

Labelled pages → Recognition models → Stored glyph profiles

Models support recognition. Profiles supply reproduction.

READ

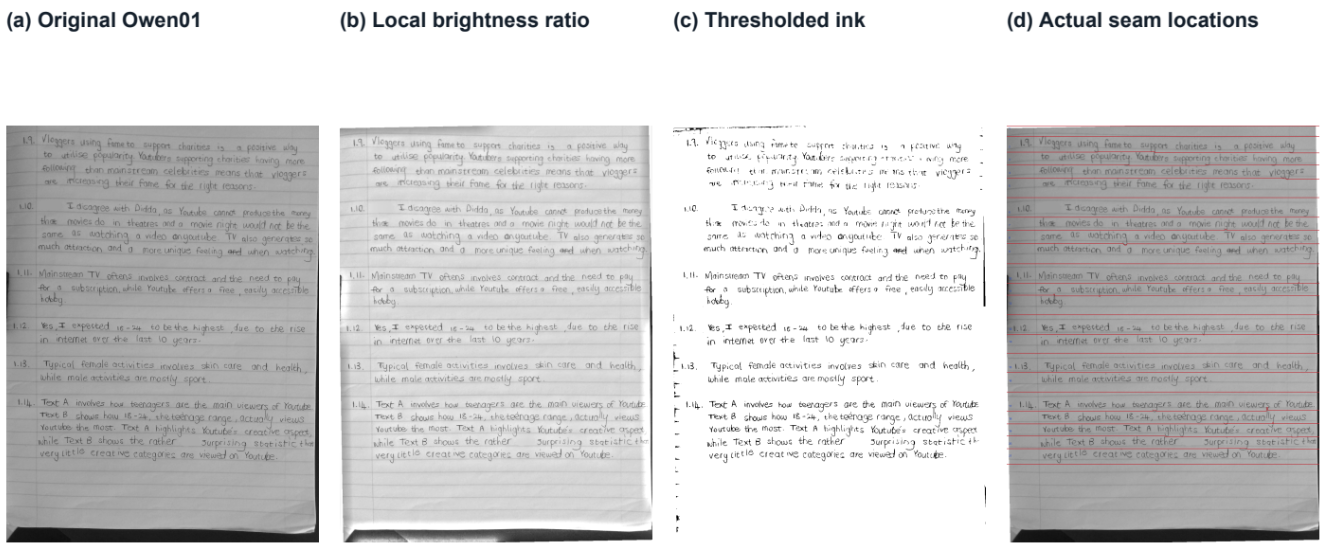
Camera photograph → Condition and split lines → Text and author

REPRODUCE

New text + profile → Glyph paths and G-code → Gantry and written page

Preparing a page for recognition

Lighting, rotation and line ownership affect everything downstream



Native input: 2448 x 3264; 20 crops; deskew +1.0 deg.

Output line 1

1.9. Vloggers using fame to support charities is a positive way

Output line 2

to utilise popularity. Youtubers supporting charities having more

Correct lighting

Compare ink with its local background.

Recover the lines

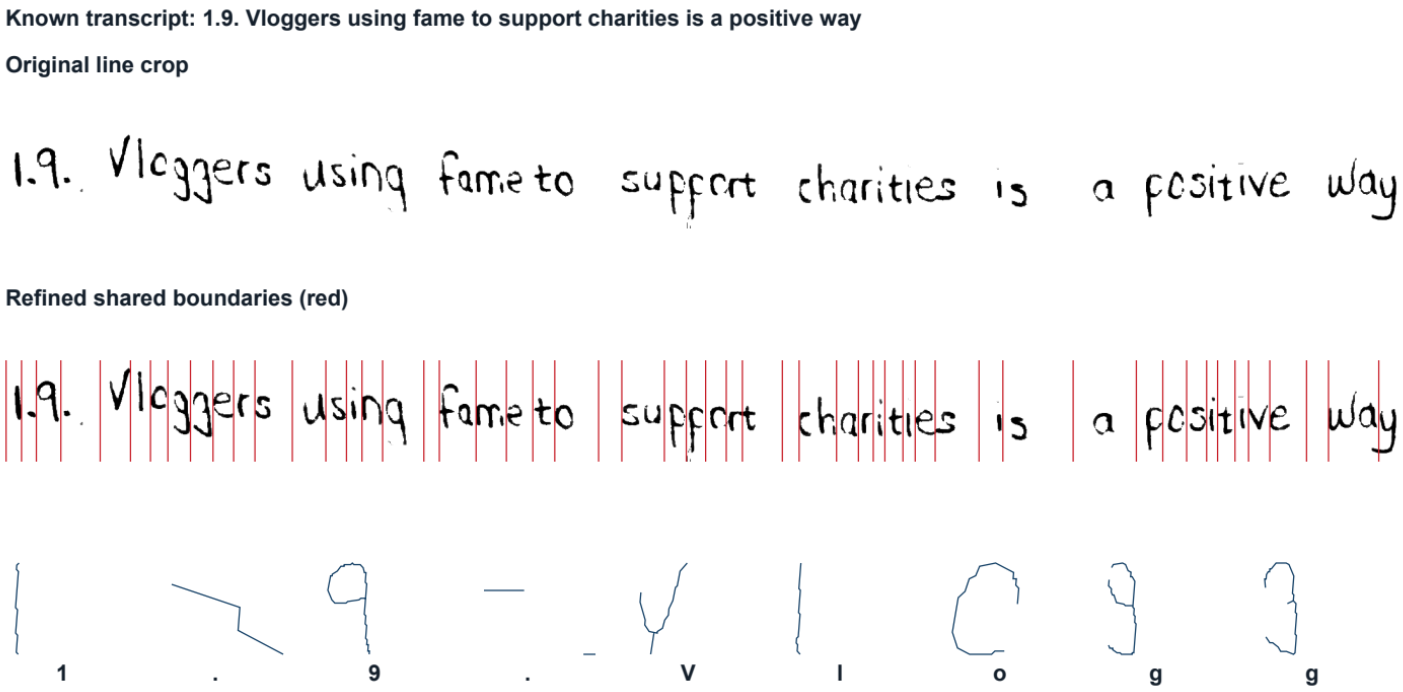
Keep neighbouring strokes in the correct crop.

Inspect the stages

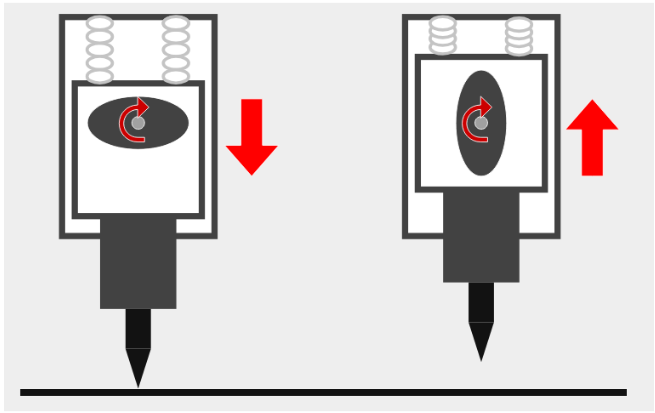
Trace a missing stroke to the operation that lost it.

Reusing handwriting as pen trajectories

Known text locates glyphs. Stored paths retain their shapes.



Aligned source line and extracted glyphs

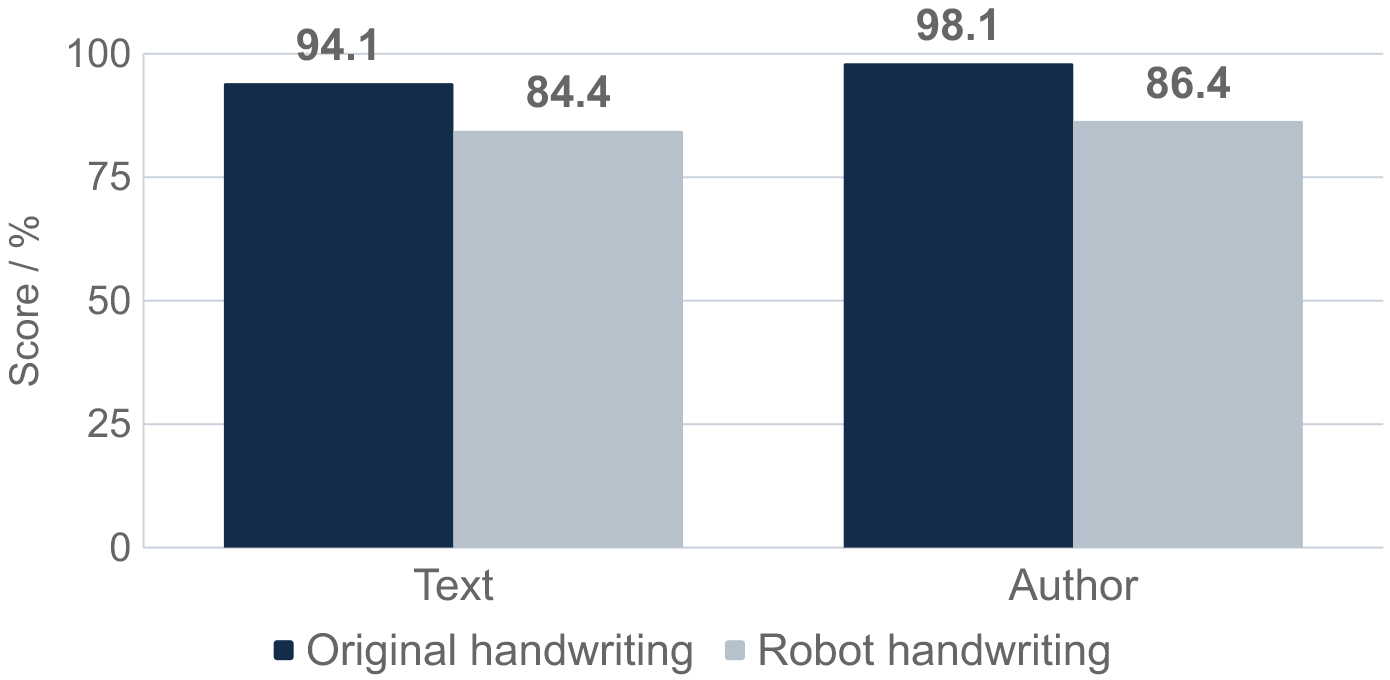


Independent pen lift

Known transcript → **Glyph paths** → **Placement and joins** → **G-code and motion**

Measured system performance

Recognition, physical output and motion are checked separately



Original targets are 95% for text and author recognition.

11

stored styles in physical output

70 / 70

lengths within 0.5 mm
Maximum error 0.49 mm

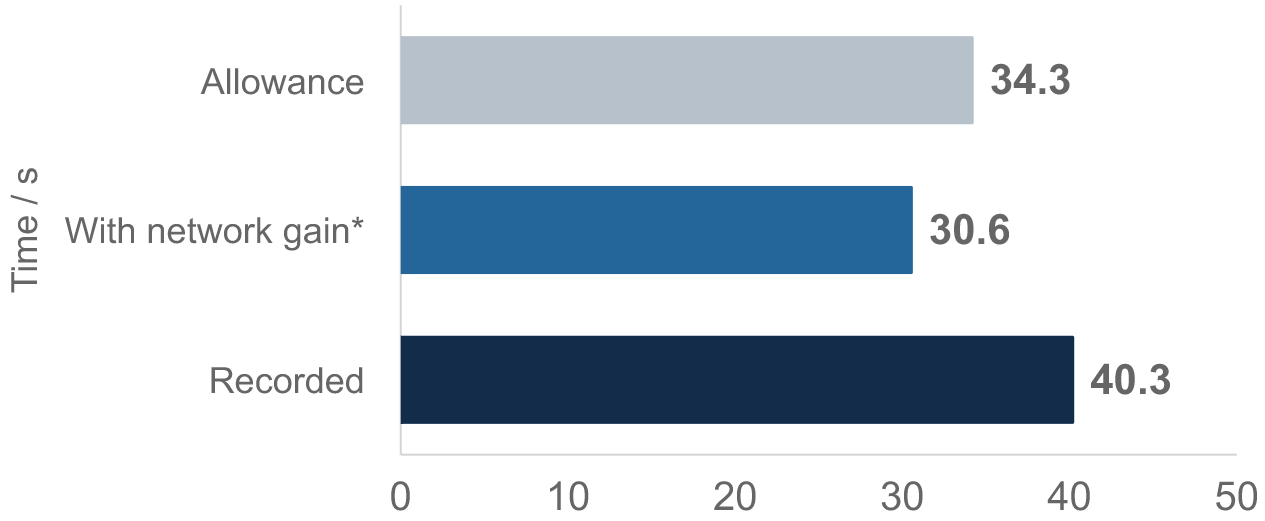
25 / 25

completed jobs within 4 s per character

Timing limit and live demonstration

Faster inference helps, while image preparation still limits some pages

Laptop example, 1291 characters



*Conditional estimate using a measured 2.76-fold network gain.
This is not a measured complete-request pass.

Demonstration sequence

01 Capture and identify

02 Select a writer and new text

03 Preview and draw

04 Compare with the original